National University of Computer and Emerging Sciences

School of Computing Fall 2023 Islamabad Campus

Question # 1[16 marks]

(a) Two basis of $\mathbb{P}_1$ are given by (2+4)

$$\mathbb{B} = \{1+x, 1-x\}, \quad \mathbb{C} = \{1, 1+x\}$$

(i) Find the coordinate vector of $p(x) = 1 + 2x$ with respect to $\mathbb{B}$ basis

(ii) Find change of basis matrix $P_{\mathbb{B}\leftarrow\mathbb{C}}$.

i) $a_1\begin{pmatrix}1\\1\end{pmatrix} + a_2\begin{pmatrix}1\\-1\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix}$

$\begin{pmatrix}1 & 1 & 1\\1 & -1 & 2\end{pmatrix} \overset{R}{\sim} \begin{pmatrix}1 & 1 & 1\\0 & -2 & 1\end{pmatrix} \Rightarrow \boxed{a_2 = -1/2}$

$a_1 = 1 + 1/2 = 3/2$

$\boxed{a_1 = \frac{3}{2}}$

$[p(x)]_B = \begin{pmatrix}3/2\\-1/2\end{pmatrix}$

ii) $P_{B\leftarrow C}$

$\left(\begin{array}{cc|cc}1 & 1 & 1 & 1\\1 & -1 & 0 & 1\end{array}\right) \overset{R}{\sim} \left(\begin{array}{cc|cc}1 & 1 & 1 & 1\\0 & -2 & -1 & 0\end{array}\right)$

$\overset{R}{\sim} \left(\begin{array}{cc|cc}1 & 1 & 1 & 1\\0 & 1 & 1/2 & 0\end{array}\right) \overset{R}{\sim} \left(\begin{array}{cc|cc}1 & 0 & 1/2 & 1\\0 & 1 & 1/2 & 0\end{array}\right)$

$P_{B\leftarrow C} = \begin{pmatrix}1/2 & 1\\1/2 & 0\end{pmatrix}$

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(b) Let $T : \mathbb{P}_2 \to \mathbb{P}_2$ defined by (5+2.5+2.5)

$$T(p(x)) = p'(x) + p(x)$$

(i) Find the matrix of transformation with respect to standard basis of $\mathbb{P}_2$ and $\mathbb{P}_1$.

(ii) Find $Kernel(T)$.

(iii) Find $Range(T)$.

i, $T(1) = 0 + 1$ $T(x) = 1 + x$, $T(x^2) = 2x + x^2$

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

ii, $AX = 0$

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right).$$

$\Rightarrow$ $\boxed{x_3 = 0}$ $x_2 = -2x_3 = 0 \Rightarrow \boxed{x_2 = 0}$

$x_1 + x_2 = 0 \Rightarrow x_1 = 0$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

$Kernel(T) = 0 + 0x + 0x^2$,

iii, $Range(T) = P_2$

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Question # 2 [21 marks]

(a) Let (12)

$$A = \begin{pmatrix} 2 & -3 & 7 \\ 0 & 5 & 1 \\ 0 & 0 & -1 \end{pmatrix}$$

Compute A^8 by using the idea of eigenvalues and eigenvectors.

Eigenvalues: 2, 5, −1 (2)

$\lambda = 2$ $(A - 2I)X = 0$

$$\begin{pmatrix} 0 & -3 & 7 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & -3 & 0 \end{pmatrix} \overset{R}{\sim} \begin{pmatrix} 0 & -3 & 7 & 0 \\ 0 & 0 & 8 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \overset{R}{\sim} \begin{pmatrix} 0 & -3 & 7 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$ (2)

$\boxed{x_3 = 0}$ $-3x_2 = 0$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} ; \quad x_1 \in \mathbb{R} \setminus \{0\}$$

$\lambda = 5$ $(A - 5I)X = 0$ (2)

$$\begin{pmatrix} -3 & -3 & 7 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -6 & 0 \end{pmatrix} \overset{R}{\sim} \begin{pmatrix} -3 & -3 & 7 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\Rightarrow \boxed{x_3 = 0}$ $-3x_1 - 3x_2 = 0$ (2)

$\Rightarrow \boxed{x_1 = -x_2}$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_2 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} ; \quad x_2 \in \mathbb{R} \setminus \{0\}$$

$\lambda = -1$ $(A + I)X = 0 \Rightarrow \begin{pmatrix} 3 & -3 & 7 & 0 \\ 0 & 6 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ $\boxed{x_2 = -\frac{1}{6}x_3}$

$3x_1 = 3\left(-\frac{1}{6}x_3\right) - 7x_3$

$3x_1 = \frac{-1-14}{2}x_3$

$x_1 = -\frac{5}{2}x_3$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_3 \begin{pmatrix} -5/2 \\ -1/6 \\ 1 \end{pmatrix} = \frac{x_3}{6} \begin{pmatrix} -15 \\ -1 \\ 6 \end{pmatrix} ; \quad x_3 \in \mathbb{R} \setminus \{0\}.$$

(2) $P = \begin{pmatrix} 1 & -1 & -15 \\ 0 & 1 & -1 \\ 0 & 0 & 6 \end{pmatrix}$, $D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -1 \end{pmatrix}$.

$D^8 = \begin{pmatrix} 2^8 & 0 & 0 \\ 0 & 5^8 & 0 \\ 0 & 0 & (-1)^8 \end{pmatrix}$

$A^8 = PD^8P^{-1}$ (2)

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(b) Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by $T(x) = Ax$ where $A = \begin{pmatrix} 0 & 1 \\ -3 & 4 \end{pmatrix}$. Find the basis $\mathbb{B}$ of $\mathbb{R}^2$ with the property that matrix of transformation $[T]_{\mathbb{B}}$ is diagonal. (5)

$\lambda^2 - 4\lambda + 3 = 0 \Rightarrow \lambda = 1, 3$

1- Eigenspace = span $\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}$

3- eigenspace = span $\left\{\begin{pmatrix} 1 \\ 3 \end{pmatrix}\right\}$

If we take basis $\mathbb{B} = \left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \end{pmatrix}\right\}$

matrix of transformation will be $\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$

(c) The eigenvalues of $A = \begin{pmatrix} 5 & -1 \\ 2 & 2 \end{pmatrix}$ are 3 and 4. Find the eigenvalues of A^3. (2)

$A = PDP^{-1}$, $D = \begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix}$.

$A^3 = PD^3P^{-1}$, $D^3 = \begin{pmatrix} 27 & 0 \\ 0 & 64 \end{pmatrix}$.

The eigenvalues of A^3 are 27 and 64.

(d) The eigenvalues of $A = \begin{pmatrix} 2 & 3 \\ x & y \end{pmatrix}$ are 4 and 8. Find the values of x and y. (2)

Sum of eigenvalues = Trace $\begin{pmatrix} 2 & 3 \\ x & y \end{pmatrix}$.

$4 + 8 = 2 + y \Rightarrow \boxed{y = 10}$

Product of eigenvalues = det(A).

$32 = 2y - 3x \Rightarrow -3x = 32 - 2(10)$

$-3x = 12$

$\boxed{x = -4}$

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Question # 3[28 marks]

(a) Let W be a subspace of $\mathbb{R}^3$ given by (4+6+5+5)

$$W = span\left\{ \begin{pmatrix}1\\0\\1\end{pmatrix}, \begin{pmatrix}0\\1\\-1\end{pmatrix} \right\}$$

(i) Find the orthogonal projection of $\mathbf{b} = \begin{pmatrix}1\\1\\1\end{pmatrix}$ on W.

(ii) Find the projection matrix that projects each vector on W.

(iii) Find orthogonal complement of W i.e., $W^{\perp}$.

(iv) Decompose $\mathbf{b}$ orthogonally as $\mathbf{b} = \mathbf{u} + \mathbf{v}$ where $\mathbf{u} \in W$ and $\mathbf{v} \in W^{\perp}$.

i) $W = span\left\{ \underset{u_1}{\begin{pmatrix}1\\0\\1\end{pmatrix}}, \underset{u_2}{\begin{pmatrix}0\\1\\-1\end{pmatrix}} \right\}$

Since $u_1 \cdot u_2 \neq 0$

By using G.S process

$v_1 = u_1 = \begin{pmatrix}1\\0\\1\end{pmatrix}$, $v_2 = u_2 - \frac{u_2 \cdot v_1}{v_1 \cdot v_1} v_1$

$= \begin{pmatrix}0\\1\\-1\end{pmatrix} - \left(-\frac{1}{2}\right)\begin{pmatrix}1\\0\\1\end{pmatrix}$

$= \begin{pmatrix}1/2\\1\\-3/2\end{pmatrix}$ or $\begin{pmatrix}1\\2\\-1\end{pmatrix}$

$W = span\left\{ \begin{pmatrix}1\\0\\1\end{pmatrix}, \begin{pmatrix}1\\2\\-1\end{pmatrix} \right\}$

$\hat{b} = \frac{b \cdot v_1}{v_1 \cdot v_1} v_1 + \frac{b \cdot v_2}{v_2 \cdot v_2} v_2$

$= \frac{2}{2}\begin{pmatrix}1\\0\\1\end{pmatrix} + \frac{2}{4}\begin{pmatrix}1\\2\\-1\end{pmatrix}$

$= \begin{pmatrix}1\\0\\1\end{pmatrix} + \begin{pmatrix}1/2\\1\\-1/2\end{pmatrix}$

$= \begin{pmatrix}3/2\\1\\1/2\end{pmatrix} \sim \begin{pmatrix}3\\2\\1\end{pmatrix}$.

The projection matrix

$P = A(A^TA)^{-1}A^T$ or

Define a map $T: \mathbb{R}^3 \to \mathbb{R}^3$

$T(x) = Proj_W^{\circ} x$

$T(e_1) = \frac{e_1 \cdot v_1}{v_1 \cdot v_1} v_1 + \frac{e_1 \cdot v_2}{v_2 \cdot v_2} v_2$

$= \frac{1}{2}\begin{pmatrix}1\\0\\1\end{pmatrix} + \frac{1}{6}\begin{pmatrix}1\\2\\-1\end{pmatrix} = \begin{pmatrix}2/3\\1/3\\1/3\end{pmatrix}$

$T(e_2) = \frac{e_2 \cdot v_1}{v_1 \cdot v_1} v_1 + \frac{e_2 \cdot v_2}{v_2 \cdot v_2} v_2$

$= 0 + \frac{2}{6}\begin{pmatrix}1\\2\\-1\end{pmatrix} = \begin{pmatrix}1/3\\2/3\\-1/3\end{pmatrix}$.

$T(e_3) = \frac{1}{2}\begin{pmatrix}1\\0\\1\end{pmatrix} + \frac{(-1)}{6}\begin{pmatrix}1\\2\\-1\end{pmatrix}$

$= \begin{pmatrix}1/3\\-1/3\\2/3\end{pmatrix}$

$P = \begin{pmatrix}2/3 & 1/3 & 1/3\\1/3 & 2/3 & -1/3\\1/3 & -1/3 & 2/3\end{pmatrix}$.

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iii) $W^{\perp} = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 0 \right.$

$\left. \text{and } \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = 0 \right\}$

$= \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : x+z=0 \text{ and } x+2y-z=0 \right\}$

$x=-z, \quad -2z+2y=0$

$y=z$

$= \left\{ \begin{pmatrix} -z \\ z \\ z \end{pmatrix} = z \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} : z \in \mathbb{R} \right\}$

iv) $b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \hat{b} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$

$b - \hat{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ 0 \end{pmatrix}$

$b = u + v = \hat{b} + (b - \hat{b})$

$= \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} + \begin{pmatrix} -2 \\ -1 \\ 0 \end{pmatrix}$

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(b) Let $c \in W^{\perp}$. Find Projection of c along W (1)

Ans: Since $c \in W^{\perp} \Rightarrow \text{proj}_W c = 0$.

(c) Let $c \in W$. Find Projection of c along W (1)

Ans:- $c \in W \Rightarrow \text{proj}_W c = c$

(d) Find a plane $y = Ax + C$ that best fits in the data (6)

$$\left\{ \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix} \begin{pmatrix}-1\\0\end{pmatrix} \begin{pmatrix}0\\-1\end{pmatrix} \right\}.$$

Solution: $y = Ax + C$.

$0 = A(1) + C$
$1 = A(0) + C$
$0 = A(-1) + C$
$-1 = A(0) + C$

$$\Rightarrow \underbrace{\begin{pmatrix}0\\1\\0\\-1\end{pmatrix}}_{b} = \underbrace{\begin{pmatrix}1 & 1\\0 & 1\\-1 & 1\\0 & 1\end{pmatrix}}_{M} \underbrace{\begin{pmatrix}A\\C\end{pmatrix}}_{X}.$$

We solve $MX = b$, by using Least squares

$$M^T M X = M^T b \quad \text{— (i)}$$

$$M^T M = \begin{pmatrix}1 & 0 & -1 & 0\\1 & 1 & 1 & 1\end{pmatrix}\begin{pmatrix}1 & 1\\0 & 1\\-1 & 1\\0 & 1\end{pmatrix} = \begin{pmatrix}2 & 0\\0 & 4\end{pmatrix}.$$

$$M^T b = \begin{pmatrix}1 & 0 & -1 & 0\\1 & 1 & 1 & 1\end{pmatrix}\begin{pmatrix}0\\1\\0\\-1\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix}.$$

$$\text{i)} \Rightarrow X = (M^T M)^{-1} M^T b = 0 = \begin{pmatrix}A\\C\end{pmatrix}.$$

$\Rightarrow y = 0$ is the best fit line.